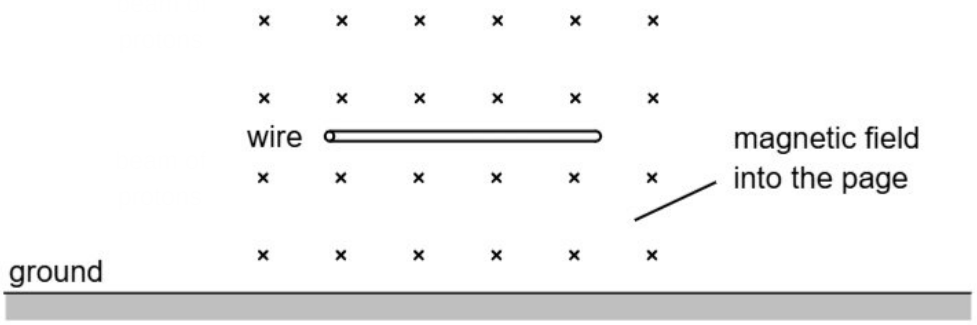
24 A current-carrying thin wire 30 cm long with a mass of 2.4 g is initially suspended at rest above the ground when placed in a uniform magnetic field of flux density 0.040 T.

The wire experiences an upward acceleration of 1.2 m s^{-2} when a current I passes through it.



If the current in the wire is $\frac{I}{2}$, what will its acceleration be?

	magnitude	direction
A	0.60 m s^{-2}	upward
B	4.3 m s^{-2}	upward
C	4.3 m s^{-2}	downward
D	5.5 m s^{-2}	downward